
Calculators and communication devices are not allowed

Answer the following questions

1. (a) Let $W = \{(a, b, c, d) : 2b = 5c + d, 4c = 2b + 2d\}$ be a subset of $\mathbb{R}^4$.
 - (i) (5 pts.) Show that W is a subspace of $\mathbb{R}^4$.
 - (ii) (5 pts.) Find a basis for W .
- (b) (5 pts.) Show that similar matrices have the same characteristic polynomials.
2. (a) (5 pts.) Write a vector equation and parametric equations for the x -axis in $\mathbb{R}^3$.
- (b) (5 pts.) Show that $U \cdot X = 0$ for every $X \in \mathbb{R}^n$ if and only if $U = O$.
- (c) (10 pts.) Show that a linear system has zero, one or infinitely many solutions.
- (d) (10 pts.) Let X and Y be nonzero vectors in $\mathbb{R}^n$. Show that X and Y are parallel, if and only if $X = cY$ for some scalar c .
3. (a) Let $S = \{X_1, X_2, \dots, X_k\}$ be a set of vectors in a vector space V .
 - (i) (5 pts.) Show that $W = \text{span}(S)$ is a subspace of V .
 - (ii) (5 pts.) Show that S is contained in W .
- (b) (5 pts.) Find the point of intersection of the line $L : \frac{x-1}{-1} = \frac{y+1}{2} = z-1$ and the plane $2x - 3y - z + 14 = 0$.
4. (a) Let $P_1(1, 0, -1), P_2(1, -1, 0), P_3(-1, -1, -1)$ be points in $\mathbb{R}^3$.
 - (i) (5 pts.) Determine whether $\overrightarrow{P_1P_2}, \overrightarrow{P_1P_3}$ and $\overrightarrow{P_2P_3}$ are basis for $\mathbb{R}^3$.
 - (ii) (5 pts.) Find an equation of the plane π , which contains the points P_1, P_2 and P_3 .
- (b) (5 pts.) Let V be a nonzero vector in $\mathbb{R}^n$ and $W = \{X \in \mathbb{R}^n : X \text{ is orthogonal to } V\}$. Show that W is a subspace of $\mathbb{R}^n$.
5. (10 pts.) Let $A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{bmatrix}$. Find a diagonal matrix D that is similar to A .

6. In each of the following questions, select the correct answer. (3 pts. each)

- (a) Let X, Y, Z be nonzero vectors in $\mathbb{R}^2$, and suppose A is a 2×2 matrix such that $AX = O$, $AY = -2Y$ and $AZ = Z$, then
- (A) $-2, 0, 1$ are eigenvalues of A .
 - (B) A is nonsingular
 - (C) X and Y are linearly dependent
 - (D) None of the above
- (b) If W_1 and W_2 are subspaces of a vector space V , then
- (A) $W_1 \cup W_2$ could be sometimes a subspace of V .
 - (B) $W_1 \cap W_2$ could be sometimes a subspace of V .
 - (C) All of the above
 - (D) None of the above
- (c) Let A and B be any $n \times n$ matrices, then
- (A) $AB = BA$
 - (B) $AB \neq BA$
 - (C) $(AB)^T = (BA)^T$
 - (D) None of the above
- (d) Consider the matrices $A = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$, $B = \begin{bmatrix} -6 & 0 \\ 0 & -1 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Then
- (A) A and B are similar matrices
 - (B) A and C are similar matrices
 - (C) B and C are similar matrices
 - (D) None of the above
- (e) Let A and B be $n \times n$ nonsingular matrices such that $A^T A^2 \text{adj}(A) - A = BA$, then
- (A) $B = A^T A \text{adj}(A) - I_n$
 - (B) $A = \frac{1}{\det(A)} (B^T + I)$
 - (C) $B = A^T A^2 \text{adj}(A^T) - I_n$
 - (D) None of the above

1. (a) (i) $2b = 5c + d, 4c = 2b + 2d \implies 2b - 5c - d = 0, 2b - 4c + 2d = 0$. W is the solution set of $AX = O$, where $A = \begin{bmatrix} 0 & -2 & 5 & 1 \\ 0 & 2 & -4 & 2 \end{bmatrix} \equiv \dots$ which is a subspace of $\mathbb{R}^4$.
(ii) $A \equiv \dots \equiv \begin{bmatrix} 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{bmatrix} \cdot x_1 = r, x_2 = -7s, x_3 = -3s, x_4 = s; r, s \in \mathbb{R} \implies (x_1, x_2, x_3, x_4) = r(1, 0, 0, 0) + s(0, -7, -3, 1) \implies B = \{(1, 0, 0, 0), (0, -7, -3, 1)\}$ is basis for W .
- (b) Let A be similar to B . Then $A = X^{-1}BX$ for some nonsingular matrix X . The characteristic polynomial of $A \equiv \det(\lambda I_n - A) = \det(\lambda X^{-1}I_n X - X^{-1}BX) = \det(X^{-1}(\lambda I_n - B)X) = \det(X^{-1}) \det(\lambda I_n - B) \det(X) = \det(\lambda I_n - B) \equiv$ The characteristic polynomial of B .
2. (a) The vector equation for the x -axis in $\mathbb{R}^3$ is $X = (0, 0, 0) + t(1, 0, 0), -\infty < t < \infty$. Parametric equations of the x -axis are $x = t, y = 0, z = 0, -\infty < t < \infty$.
- (b) (1) Let $X = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$. If $U = O = (0, 0, \dots, 0)$, then $U \cdot X = 0x_1 + 0x_2 + \dots + 0x_n = 0$.
(2) In particular $U \cdot U = 0 \implies \|U\|^2 = 0 \implies \|U\| = 0 \implies U = O$.
- (c) Let X_1 and X_2 be different solutions for $AX = B$. Then $X_0 = X_1 - X_2 \neq O$ is a solution for $AX = O$. Let $Y = X_1 + sX_0$. Then Y is a solution for $AX = B$ for every $s \in \mathbb{R}$.
- (d) (5 pts.) (1) Let $X = cY \implies \cos \theta = \frac{(cY) \cdot Y}{\|cY\| \|Y\|} = \frac{c(Y \cdot Y)}{|c| \|Y\|^2} = \frac{c}{|c|} = \pm 1 \implies \theta = 0 \text{ or } \theta = \pi \implies X \text{ and } Y \text{ are parallel.}$
(5 pts.) (2) If X and Y are parallel. Let U_1 and U_2 be unit vectors in the directions of X and Y , respectively $\implies U_1 = \frac{1}{\|X\|}X, U_2 = \frac{1}{\|Y\|}Y$ and either $U_1 = U_2$ or $U_1 = -U_2 \implies$ either $X = \|X\| U_2$ or $X = -\|X\| U_2 \implies$ either $X = \|X\| \frac{1}{\|Y\|}Y = cY$ or $X = -\|X\| \frac{1}{\|Y\|}Y = -cY$, where $c = \frac{\|X\|}{\|Y\|}$.
3. (a) Let $S = \{X_1, X_2, \dots, X_k\}$ be a set of vectors in a vector space V and $W = \text{span}(S)$.
(i) (1) Since $O = 0X_1 + 0X_2 + \dots + 0X_k \implies O \in W \implies W \neq \emptyset$.
 (α) Let $X = c_1X_1 + c_2X_2 + \dots + c_kX_k$ and $Y = d_1X_1 + d_2X_2 + \dots + d_kX_k \in W \implies X + Y = (c_1 + d_1)X_1 + (c_2 + d_2)X_2 + \dots + (c_k + d_k)X_k \in W$
 (β) Let $t \in \mathbb{R}$ and $X = c_1X_1 + c_2X_2 + \dots + c_kX_k \implies tX = (tc_1)X_1 + (tc_2)X_2 + \dots + (tc_k)X_k \in W$
(ii) For every $i = 1, 2, \dots, k$, we have $X_i = 0X_1 + \dots + 0X_{i-1} + 1X_i + 0X_{i+1} + \dots + 0X_k \implies \forall X_i \in S$ we have $X_i \in W \implies S \subseteq W$.
- (b) The parametric equations of the line L are: $x = 1 - t, y = -1 + 2t, z = 1 + t \implies 2(1 - t) - 3(-1 + 2t) - (1 + t) + 14 = 0 \implies 18 - 9t = 0 \implies t = 2$.
The point of intersection is $(-1, 3, 3)$.

4. (a) (i) $\overrightarrow{P_1P_2} = (0, -1, 1)$, $\overrightarrow{P_1P_3} = (-2, -1, 0)$ and $\overrightarrow{P_2P_3} = (-2, 0, -1)$. Since $\begin{vmatrix} 0 & -1 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{vmatrix} =$

0, then $\overrightarrow{P_1P_2}$, $\overrightarrow{P_1P_3}$ and $\overrightarrow{P_2P_3}$ are not basis. (They are Linearly dependent and do not form a spanning set for $\mathbb{R}^3$).

(ii) $N = \overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} i & j & k \\ 0 & -1 & 1 \\ -2 & -1 & 0 \end{vmatrix} = (1, -2, -2)$ is normal to the plane.

Equation of the plane is: $(1)(x-1) + (-2)(y-0) + (-2)(z+1) = 0 \implies x - 2y - 2z - 3 = 0$

(b) (1) Since $O \cdot V = 0$, then $O \in W$ and hence, $W \neq \emptyset$. $(\alpha) \forall X, Y \in W, X \cdot V = 0, Y \cdot V = 0$ and since $(X+Y) \cdot V = X \cdot V + Y \cdot V = 0+0=0$, then $X+Y \in W$.

(β) Let $X \in W$ and $c \in \mathbb{R}$, then $X \cdot V = 0$. Since $(cX) \cdot V = c(X \cdot V) = c(0) = 0$, then $(cX) \in W$.

5. The characteristic polynomial of A is $|\lambda I_3 - A| = \begin{vmatrix} \lambda+1 & 0 & 0 \\ 0 & \lambda-1 & -2 \\ -1 & 0 & \lambda \end{vmatrix} = \lambda(\lambda-1)(\lambda+1) \implies \lambda_1 = -1, \lambda_2 = 0, \lambda_3 = 1$ are the eigenvalues of A .

(i) Solve $((1)I_3 - A)X = O \implies \left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ -1 & 0 & 1 & 0 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \implies$

$X = r \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, r \neq 0$ are the eigenvectors corresponding to the eigenvalue $\lambda_1 = 1$

(ii) Solve $((0)I_3 - A)X = O \implies \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & -1 & -2 & 0 \\ -1 & 0 & 0 & 0 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \implies$

$X = r \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}, r \neq 0$ are the eigenvectors corresponding to the eigenvalue $\lambda_2 = 0$

(iii) Solve $((-1)I_3 - A)X = O \implies \left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & -2 & -2 & 0 \\ -1 & 0 & -1 & 0 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \implies$

$X = r \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, r \neq 0$ are the eigenvectors corresponding to the eigenvalue $\lambda_3 = -1$.

Let $P = \begin{bmatrix} 0 & 0 & -1 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is the diagonal matrix similar to A , since $P^{-1}AP = D$.

6. (a) Let X, Y, Z be nonzero vectors in $\mathbb{R}^2$, and suppose A is a 2×2 matrix, such that $AX = O$, $AY = -2Y$ and $AZ = Z$, then
 (D) None of the above
- (b) If W_1 and W_2 are subspaces of a vector space V , then
 (A) $W_1 \cup W_2$ could be sometimes a subspace of V .
- (c) Let A and B be any $n \times n$ matrices, then
 (D) None of the above
- (d) Consider the matrices $A = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$, $B = \begin{bmatrix} -6 & 0 \\ 0 & -1 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.
 (D) None of the above
- (e) Let A and B be $n \times n$ nonsingular matrices such that $A^T A^2 \text{adj}(A) - A = BA$, then
 (B) $A = \frac{1}{\det(A)} (B^T + I)$